

Numerical Integration Using the Trapezoidal Rule

1. Introduction

Numerical integration is an important topic in numerical analysis used to approximate definite integrals when analytical solutions are difficult or impossible to obtain. One of the simplest and most widely used numerical methods is the Trapezoidal Rule. This method approximates the area under a curve by dividing it into trapezoids instead of rectangles.

2. Theoretical Background

The Trapezoidal Rule is based on approximating the function by a straight line between two points. The area under the curve between a and b is approximated by the area of a trapezoid.

For single interval:

$$\int_a^b f(x)dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

For multiple intervals (Composite Trapezoidal Rule):

$$h = \frac{b-a}{n}$$
$$\int_a^b f(x)dx \approx \frac{h}{2} [f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)]$$

3. Algorithm:

To apply the Trapezoidal Rule:

1. Choose the number of subintervals n .
2. Compute step size $h = \frac{b-a}{n}$.
3. Compute the function values at each point.
4. Apply the composite trapezoidal formula.
5. Multiply the final summation by $h/2$.

4. Example

Approximate:

$$\int_0^2 x^2 dx$$

Let $n = 4$

Step size:

$$h = \frac{2 - 0}{4} = 0.5$$

$$x_0 = 0, x_1 = 0.5, x_2 = 1, x_3 = 1.5, x_4 = 2$$

$$f(0.5) = 0.25, f(1) = 1, f(1.5) = 2.25, f(2) = 4$$

Using the formula:

$$\int_0^2 x^2 dx = \frac{0.5}{2} [0 + 2(0.25) + 2(1) + 2(2.25) + 4] = 2.75$$

Exact value:

$$\frac{8}{3} \approx 2.67$$

The approximation is close to the exact solution

6. Applications

The Trapezoidal Rule is used in:

- Engineering calculations
- Physics problems
- Signal processing
- Computer simulations
- When experimental data is given in table form

7. Conclusion

The Trapezoidal Rule is a simple and effective numerical integration technique. Although it may not be as accurate as Simpson's Rule, it is easy to implement and works well for smooth functions.